



Progranulin analysis of plasma and CSF samples from PPMI

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Summary:

Plasma and CSF samples from PPMI were analyzed by using an MSD assay in order to determine human progranulin levels both in the blood and CSF of healthy controls and Parkinson's disease (PD) patients, including those who carry PD-associated mutations in *LRRK2* and/or *GBA*.

Revision 1. Version 1 contained the methodology of plasma sample processing and analysis. The current version includes the methodology of CSF sample processing and analysis.

Materials:

- 1) MSD 96-well Standard Uncoated Plates (MSD, Cat #L15XA-1)
- 2) Recombinant Human Progranulin (Cat No. CRH026B, Cell Sciences) Store at <-60°C
- 3) Ruthenylated goat polyclonal anti-HuPGRN (Cat No. AF2420, R&D Systems) (Detection antibody) in PBS, prepared by using Standard MSD protocol
- 4) Unconjugated goat polyclonal anti-HuPGRN (Cat No. AF2420, R&D Systems) (Capture antibody) in PBS. Store at ≤ -60°C for long term or at 4°C for short-term
- 5) 1XPBS (various vendors)
- 6) MSD Blocker A Kit (Cat No. R93AA, Meso Scale Discovery)
- 7) Read Buffer T (4X) (MSD, Cat No R92TC-2)
- 8) Deionized (DI) Water (Gibco)
- 9) 20X PBS-Tween-20 Solution (Cat No. 28352, Thermo Fisher)





10) Wash Buffer: 1X PBS + 0.05% Tween-20

11) Blocking Buffer: 5% MSD Blocker A in 1XPBS

12) Assay Diluent: 1% MSD Blocker A in 1XPBS, 0.05% Tween-20

Method:

On the day prior to running the assay coat an MSD 96-well Standard Uncoated Plates with capture antibody diluted in PBS. Seal and incubate the plate O/N.

On the day of the assay, wash the plate with wash buffer then add blocking buffer. While the plate is blocking prepare samples and standards. Spin down plasma samples at 3500xg for 10 minutes at 4°C. Transfer plasma and/or CSF samples into a 96 well V-bottom plate and dilute as needed in assay diluent. Prepare a standard curve as needed by serially diluting the recombinant human progranulin protein in assay diluent.

After the blocking step is complete, wash the MSD 96-well plate and add the prepared standards and plasma and/or CSF samples and incubate at RT. After the incubation step, wash the plate and add the detection antibody diluted in assay buffer and incubate at RT. When done incubating, wash the plate and add read buffer and read plate on MSD Sector Imager 600 within 20 minutes.

Analysis

The plasma samples were organized into 8 blocks analyzed on two separate run dates (batches). Within each of the blocks 6 pooled sample QCs (quality control samples) were analyzed 6 times. High, medium, and low QCs were also included to monitor assay performance.

The CSF samples were organized into 8 blocks, all of which were run on the same day. Within each of the blocks 6 pooled sample QCs (quality control samples) were analyzed 6 times.

For both plasma and CSF, samples were run in technical duplicates and the average of the two data points was reported. In the CSF analysis, block 3 did not meet standard acceptance criteria because the technical replicates had a greater than 20% CV in a significant number of samples; therefore, this block was re-run. The re-run was within acceptance criteria so the data from the re-run were reported.

Data was analyzed through the Prism software. All raw data was interpolated to the standard curve using a sigmoidal 4 parameter fit with $1/y^2$ weighting for curve fitting. For plasma, batch to batch variations were corrected by normalizing the individual sample data by the median fold change of pooled samples per each of the 8 blocks to derive the final adjusted concentration values. No block-to-block normalization was performed for the CSF results.





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Data Availability

Analyte measurements from all study samples have been provided. Measured values in the 6 pooled quality control samples are available upon request.

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